

The Collected Mathematical Papers of Arthur Cayley

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Papers [81] (cont.), [82], [83], [84], [85], [86] (part)

[81] Addition au Mémoire sur quelques transmutations des lignes courbes (*continued*)

[PDF p. 501 / Book p. 479]

celle des cercles de se transmuter en des ovales de Descartes (seulement M. Chasles paraît ne pas avoir remarqué que ces ovales étaient nécessairement des limaçons), et c'est la lecture de cette Note qui m'a appris cette théorie de la transmutation des cercles.

En supposant que la conique à transmuter passe par les trois points P, Q, R , nous aurons

$$\alpha + \delta = 0, \quad \beta + \delta = 0, \quad \gamma + \delta = 0;$$

les points doubles, identiques (dans ce cas) avec les points P, Q, R , deviennent des points de rebroussement, et les droites QR, RP, PQ , au lieu d'être des tangentes doubles, sont tout simplement les droites qui passent chacune par deux points de rebroussement. Ajoutons que les tangentes de la courbe, aux trois points de rebroussement respectivement, sont les droites

$$x - w = 0, \quad y - w = 0, \quad z - w = 0.$$

Il y a encore un cas particulier à considérer, savoir celui où la conique à transmuter est telle que, par rapport à cette conique, les points Q et R sont situés chacun dans la polaire de l'autre; on a alors $f = 0$, cas qui échappe à l'analyse ci-devant employée. On voit sans peine que les deux points doubles ($y = w, x = z$), ($z = w, y = z$) deviennent ici identiques, ce qui donne lieu à un point d'osculation. La droite QR et la droite $w = 0$ ne sont plus des tangentes doubles proprement dites, mais ces droites deviennent l'une et l'autre identiques avec la tangente au point d'osculation.

Note sur les ovales de Descartes.

De l'équation de ces ovales,

$$\sqrt{(x-a)^2 + y^2} = m\sqrt{x^2 + y^2} + n,$$

on tire d'abord

$$(1 - m^2)(x^2 + y^2) - 2ax + a^2 - n^2 = 2mn\sqrt{x^2 + y^2},$$

puis

$$\begin{aligned} (1 - m^2)^2(x^2 + y^2)^2 - 4a(1 - m^2)x(x^2 + y^2) \\ + 2\{a^2(1 - m^2) - n^2(1 + m^2)\}(x^2 + y^2) + 4a^2x^2 \\ - 4a(a^2 - n^2)x + (a^2 - n^2)^2 = 0. \end{aligned}$$

Pour trouver la nature de la courbe à l'infini, mettons $x + yi = \xi$, $x - yi = \eta$, $i = \sqrt{-1}$, et introduisons la quantité ζ de manière à rendre l'équation homogène. Cela donne

$$\begin{aligned} (1 - m^2)^2\xi^2\eta^2 - 2a(1 - m^2)(\xi + \eta)\xi\eta\zeta \\ + 2\{a^2(1 - m^2) - n^2(1 + m^2)\}\xi\eta\zeta^2 + a^2(\xi + \eta)^2\zeta^2 \\ - 2a(a^2 - n^2)(\xi + \eta)\zeta^3 + (a^2 - n^2)^2\zeta^4 = 0; \end{aligned}$$

ce qui fait voir, sans la moindre peine, qu'il y a des points de rebroussement aux points $(\xi = 0, \zeta = 0)$, $(\eta = 0, \zeta = 0)$, savoir, aux points où l'infini, considéré comme droite, est rencontré par les droites $x + yi = 0$, $x - yi = 0$, qui sont les axes imaginaires du point $x = 0, y = 0$.

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Nous pouvons remarquer, en passant, que l'équation

$$(1 - m^2)(x^2 + y^2) - 2ax + (a^2 - n^2) = 2mn\sqrt{x^2 + y^2},$$

conduit, avec beaucoup de facilité, à une autre propriété, donnée par M. Chasles dans la Note déjà citée. En effet, en mettant

$$a' = \frac{n^2 - a^2}{a(m^2 - 1)}, \quad m' = \frac{m(n^2 - 1)}{a(m^2 - 1)}, \quad n' = \frac{n}{a},$$

cette équation se transforme en

$$(1 - m'^2)(x^2 + y^2) - 2a'x + (a'^2 - n'^2) = 2m'n'\sqrt{x^2 + y^2},$$

et par là on voit que l'équation primitive peut se transformer en

$$\sqrt{(x - a')^2 + y^2} = m'\sqrt{x^2 + y^2} + n',$$

c'est-à-dire, il y a toujours un troisième foyer de la courbe.

Il ne reste qu'à démontrer que la transformée (selon la méthode de M. Chasles) d'un cercle est toujours un limaçon. Soit, pour cela,

$$r^2 - 2ar \cos \theta + \delta = 0$$

l'équation du cercle; en mettant $\sqrt{mr}$ au lieu de r , et $\frac{1}{2}\theta$ au lieu de θ , cette équation devient

$$mr - 2a\sqrt{mr} \cos \frac{1}{2}\theta + \delta = 0,$$

ce qui donne $(mr + \delta)^2 = 2a^2mr(1 + \cos \theta)$, ou, en mettant $r \cos \theta = x$ et en réduisant à une forme rationnelle,

$$\{m^2(x^2 + y^2) + \delta^2 - 2a^2mx\}^2 - 4a^2m^2(\delta - a^2)(x^2 + y^2) = 0,$$

ce qui appartient évidemment à un ovale de Descartes. En mettant $y = 0$, l'équation devient

$$\{m^2x^2 + 2m(\delta - 2a^2)x + \delta^2\}(mx - \delta)^2 = 0,$$

c'est-à-dire le point $mx - \delta = 0$, $y = 0$ est point double, ou la courbe est le limaçon de Pascal.

[82] On the Triadic Arrangements of Seven and Fifteen Things

[From the *Philosophical Magazine*, vol. xxxvii. (1850), pp. 50–53.]

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There is no difficulty in forming with seven letters, a, b, c, d, e, f, g , a system of seven triads containing every possible duad; or, in other words, such that no two triads of the system contain the same duad. One such system, for instance, is

$$abc, \quad ade, \quad afg, \quad bdf, \quad beg, \quad cdg, \quad cef;$$

and this is obviously one of six different systems obtained by permuting the letters a, b, c . We have therefore six different systems containing the triad abc ; and there being the same number of systems containing the triads abd , abe , abf and abg respectively, there are in all thirty-five different systems, each of them containing every possible duad. It is deserving of notice, that it is impossible to arrange the thirty-five triads formed with the seven letters into five systems, each of them possessing the property in question. In fact, if this could be done, the system just given might be taken for one of the systems of seven triads. With this system we might (of the systems of seven triads which contain the triad abd) combine either the system

$$abd, \quad acg, \quad aef, \quad bce, \quad bfg, \quad dcf, \quad deg,$$

or the system

$$abd, \quad acf, \quad aeg, \quad bcg, \quad bef, \quad dce, \quad dfg;$$

(but any one of the other *abd* systems would be found to contain a triad in common with the given *abc* system, and therefore cannot be made use of: for instance, the system *abd, acg, aef, bcf, beg, dce, dfg* contains the triad *beg* in common with the given *abc* system): and whichever of the two proper *abd* systems we select to combine with the given *abc* system, it will be found that there is no *abe* system which does not contain some triad in common, either with the *abc* system or with the *abd* system.

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The order of the letters in a triad has been thus far disregarded. There are some properties which depend upon considering the triads obtained by cyclical permutations of the three letters as identical, but distinct from the triads obtained by a permutation of two letters, or inversion. Thus *abc, bca, cab* are to be considered as identical *inter se*, but distinct from the triads *acb, cba, bac*, which are also identical *inter se*. I write down the system (equivalent, as far as the mere combination of the letters is concerned, to the system at the commencement of this paper)

$$ade, \quad afg, \quad bdf, \quad bge, \quad cdg, \quad cef, \quad cba,$$

derived, it is to be observed, from a pair of triads, *ade, afg*, by a cyclical permutation of the *c, f, g*, and by successively changing the *a* into *b* and into *c*, the remaining triad of the system being the letters *a, b, c* taken in an inverse order. Let it be proposed to derive the system in the same manner from any other two triads of the system; for instance, from the triads *acb, ade*. The process of derivation gives

$$acb, \quad ade, \quad gcd, \quad geb, \quad fce, \quad fbd, \quad fga, \tag{1}$$

which is, in fact, the original system. But attempt to derive the system from the two triads *ebg, efc*, the process of derivation gives

$$ebg, \quad efc, \quad dbf, \quad dcg, \quad abc, \quad agf, \quad ade,$$

which is not the original system, inasmuch as the triads *dbf, dcg, abc, agf* are inversions of the triads *bdf, cdg, cba, afg* of the original system. The point to be attended to, however, is, that both triads of the pair *dbf, dcg*, or of the pair *abc, agf*, are inversions of the triads of the corresponding pair in the original system; the pair is either reproduced (as the pair *efc, dbf*), or there is an inversion of both triads. Where there is no such inversion of the triads of a pair, the system may be said to be properly reproduced; and where there is inversion of the triads of one or more pairs, to be improperly reproduced. There is no difficulty in seeing that the system is properly reproduced from a pair of triads containing in common any one of the letters *a, b, c* or *d*, and improperly reproduced from pairs of triads containing in common any one of the letters *d, e* or *f*. It is owing to the reproduction, proper or improper, of the system from any pair of duads that it is possible to form a system of “octaves” analogous to the quaternions of Sir William Hamilton; the impossibility of a corresponding system of fifteen imaginary quantities arises from the circumstance of there being always, in whatever manner the system of triads is formed, an inversion of a single triad of some one or more pairs of triads containing a letter in common. When the system is considered as successively derived from different pairs, the system is not according to the previous definition reproduced either properly or improperly. A system of triads having the necessary properties with respect to the mere combination of the letters (viz. that $\alpha\beta\gamma$ and $\alpha\delta\epsilon$ being any two

Footnote. The order of the letters *f, g* is selected so as to reproduce the original system so far as the mere combination of the letters is concerned.

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triads having a letter in common, there shall be triads such as $\delta\beta\epsilon$, $\zeta\gamma\epsilon$, and $\eta\delta\epsilon$, $\zeta\eta\delta$) may easily be found; the system to be presently given of the triads of fifteen things would answer the purpose. And so would many other systems.

Dropping the consideration of the order of the letters which form a triad, I pass to the case of a system of fifteen letters, *a, b, c, d, e, f, g, h, i, j, k, l, m, n, o*. It is possible in this case, not only to form systems of thirty-five triads containing every possible duad, but this can be done in such manner that the system of thirty-five triads can be arranged in seven systems of five triads, each of these systems containing the

fifteen letters.¹ My solution is obtained by a process of derivation from the arrangements *ab.cf.dg.eh* and *ab.cd.ef.gh* as follows; viz. the triads are

<i>iab</i>	<i>jac</i>	<i>kaf</i>	<i>lad</i>	<i>mag</i>	<i>nae</i>	<i>oah</i>
<i>icf</i>	<i>jfb</i>	<i>kbc</i>	<i>lce</i>	<i>mch</i>	<i>nod</i>	<i>ocg</i>
<i>idg</i>	<i>jde</i>	<i>kdh</i>	<i>lgb</i>	<i>mbd</i>	<i>ngf</i>	<i>ofd</i>
<i>ieh</i>	<i>jhg</i>	<i>kge</i>	<i>lhf</i>	<i>mfe</i>	<i>nhb</i>	<i>obe</i>

and a system formed with *i, j, k, l, m, n, o*, which are then arranged in the form

<i>klo</i>	<i>ino</i>	<i>jmo</i>	<i>ilm</i>	<i>jln</i>	<i>ijk</i>	<i>kmn</i>
<i>iab</i>	<i>jac</i>	<i>lad</i>	<i>nae</i>	<i>kaf</i>	<i>mag</i>	<i>oah</i>
<i>ncd</i>	<i>mdb</i>	<i>kbc</i>	<i>ocg</i>	<i>mch</i>	<i>lce</i>	<i>icf</i>
<i>mief</i>	<i>keg</i>	<i>ieh</i>	<i>jfb</i>	<i>obe</i>	<i>ofd</i>	<i>jde</i>
<i>jgh</i>	<i>lhf</i>	<i>nfg</i>	<i>khd</i>	<i>idg</i>	<i>nhb</i>	<i>lbg</i>

an arrangement, which, it may be remarked, contains eight different systems (such as have been considered in the former part of this paper) of seven letters such as *i, j, k, l, m, n, o*; and seven of other seven letters, such as *i, j, k, a, b, c, f*.² The theory of the arrangement seems to be worth further investigation.

Assuming that the four hundred and fifty-five triads of fifteen things can be arranged in thirteen systems of thirty-five triads, each system of thirty-five triads containing every possible duad, it seems natural to inquire whether the thirteen systems can be obtained from any one of them by cyclical permutations of thirteen letters. This is, I think, impossible. For let the cyclical permutation be of the letters *a, b, c, d, e, f, g, h, i, j, k, l, m*. Consider separately the triads which contain the letter *n* and the letter *o*; neither of these systems of triads contains the letter, whatever it is, which forms a triad with *n* and *o*. Hence, omitting the letters *n, o*, we have two different sets, each of them of six duads, and composed of the same twelve letters. And each of these systems of duads ought, by the cyclical permutation

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in question, to produce the whole system of the seventy-eight duads of the thirteen letters. Hence arranging the duads of the thirteen letters in the form

<i>ab</i>	<i>bc</i>	<i>cd</i>	<i>de</i>	<i>ef</i>	<i>fg</i>	<i>gh</i>	<i>hi</i>	<i>ij</i>	<i>jk</i>	<i>kl</i>	<i>lm</i>	<i>ma</i>
<i>ac</i>	<i>bd</i>	<i>ce</i>	<i>df</i>	<i>eg</i>	<i>fh</i>	<i>gi</i>	<i>hj</i>	<i>ik</i>	<i>jl</i>	<i>km</i>	<i>la</i>	<i>mb</i>
<i>ad</i>	<i>be</i>	<i>cf</i>	<i>dg</i>	<i>eh</i>	<i>fi</i>	<i>gj</i>	<i>hk</i>	<i>il</i>	<i>jm</i>	<i>ka</i>	<i>lb</i>	<i>mc</i>
<i>ae</i>	<i>bf</i>	<i>cg</i>	<i>dh</i>	<i>ei</i>	<i>fj</i>	<i>gk</i>	<i>hl</i>	<i>im</i>	<i>ja</i>	<i>kb</i>	<i>lc</i>	<i>md</i>
<i>af</i>	<i>bg</i>	<i>ch</i>	<i>di</i>	<i>ej</i>	<i>fk</i>	<i>gl</i>	<i>hm</i>	<i>ia</i>	<i>jb</i>	<i>kc</i>	<i>ld</i>	<i>me</i>
<i>ag</i>	<i>bh</i>	<i>ci</i>	<i>dj</i>	<i>ek</i>	<i>fl</i>	<i>gm</i>	<i>ha</i>	<i>ib</i>	<i>jc</i>	<i>kd</i>	<i>le</i>	<i>mf</i>

and consequently the duads of each set ought to be situated one duad in each line. Suppose the sets of duads are composed of the letters *a, b, c, d, e, f, g, h, i, j, k, l*, it does not appear that there is any set of six duads composed of these letters, and situated one duad in each line, other than the single set *al, bk, cj, di, eh, fg*; and the same being the case for any twelve letters out of the thirteen, the derivation of the thirteen systems of thirty-five triads by means of the cyclical permutations of thirteen letters is impossible. And there does not seem to be any obvious rule for the derivation of the thirteen systems from any one of them, or any *prima facie* reason for believing that the thirteen systems do really exist, it having been already shown that such systems do not exist in the case of seven things.

[83] On Curves of Double Curvature and Developable Surfaces

[From the *Cambridge and Dublin Mathematical Journal*, vol. v. (1850), pp. 18–22.]

¹The problem was proposed by Mr Kirkman, and has, to my knowledge, excited some attention in the form “To make a school of fifteen young ladies walk together in threes every day for a week so that each two may walk together.” It will be seen from the text that I am uncertain as to the existence of a solution to the further problem suggested by Mr Sylvester, “to make the school walk every week in the quarter so that each three may walk together.”

²I have somewhat altered this sentence so as to express more clearly what appeared to be the meaning of it.

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This is translated with some slight alterations from the Memoir in Liouville, t. x. (1845), pp. 245–250, [30].

[84] On the Developable Surfaces which arise from Two Surfaces of the Second Order

[From the *Cambridge and Dublin Mathematical Journal*, vol. v. (1850), pp. 46–57.]

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Any two surfaces considered in relation to each other give rise to a curve of intersection, or, as I shall term it, an Intersect and a circumscribed Developable³ or Envelope. The Intersect is of course the edge of regression of a certain Developable which may be termed the Intersect-Developable, the Envelope has an edge of regression which may be termed the Envelope-Curve. The order of the Intersect is the product of the orders of the two surfaces, the class of the Envelope is the product of the classes of the two surfaces. When neither the Intersect breaks up into curves of lower order, nor the Envelope into Developables of lower class, the two surfaces are said to form a proper system. In the case of two surfaces of the second order (and class) the Intersect is of the fourth order and the Envelope of the fourth class. Every proper system of two surfaces of the second order belongs to one of the following three classes: A. There is no contact between the surfaces; B. There is an ordinary contact; C. There is a singular contact. Or the three classes may be distinguished by reference to the conjugates (conjugate points or planes) of the system. A. The four conjugates are all distinct; B. Two conjugates coincide; C. Three conjugates coincide.

To explain this it is necessary to remark that in the general case of two surfaces of the second order not in contact (that is for systems of the class A) there is a certain tetrahedron such that with respect to either of the surfaces (or more generally with respect to any surface of the second order passing through the Intersect of the system

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or inscribed in the Envelope) the angles and faces of the tetrahedron are reciprocals of each other, each angle of its opposite face, and *vice versa*. The angles of the tetrahedron are termed the conjugate points of the system, and the faces of the tetrahedron are termed the conjugate planes of the system, and the term conjugates may be used to denote indifferently either the conjugate planes or the conjugate points. A conjugate plane and the conjugate point reciprocal to it are said to correspond to each other. Each conjugate point is evidently the point of intersection of the three conjugate planes to which it does not correspond, and in like manner each conjugate plane is the plane through the three conjugate points to which it does not correspond.

In the case of a system belonging to the class B, two conjugate points coincide together in the point of contact forming what may be termed a double conjugate point, and in like manner two conjugate planes coincide in the plane of contact (that is the tangent plane through the point of contact) forming what may be termed a double conjugate plane. The remaining conjugate points and planes may be distinguished as single conjugate points and single conjugate planes. It is clear that the double conjugate plane passes through the three conjugate points, and that the double conjugate point is the point of intersection of the three conjugate planes: moreover each single conjugate plane passes through the single conjugate point to which it does not correspond and the double conjugate point; and each single conjugate point lies on the line of intersection of the single conjugate plane to which it does not correspond and the double conjugate plane.

In the case of a system belonging to the class C, three conjugate points coincide together in the point of contact forming what may be termed a triple conjugate point, and three conjugate planes coincide together in the plane of contact forming a triple conjugate plane. The remaining conjugate point and conjugate

³The term ‘Developable’ is used as a substantive, as the reciprocal to ‘Curve,’ which means curve of double curvature. The same remark applies to the use of the term in the compound Intersect-Developable. For the signification of the term ‘singular contact,’ employed lower down, see Mr Salmon’s memoir ‘On the Classification of Curves of Double Curvature,’ [same volume] p. 32.

plane may be distinguished as the single conjugate point and single conjugate plane. The triple conjugate plane passes through the two conjugate points and the triple conjugate point lies on the line of intersection of the two conjugate planes; the single conjugate plane passes through the triple conjugate point and the single conjugate point lies on the triple conjugate plane.

Suppose now that it is required to find the Intersect-Developable of two surfaces of the second order. If the equations of the surfaces be $T = 0$, $T' = 0$ (T, T' being homogeneous functions of the second order of the coordinates ξ, η, ζ, ω), and x, y, z, w represent the coordinates of a point upon the required developable surface: if moreover U, U' are the same functions of x, y, z, w that T, T' are of ξ, η, ζ, ω and $X, Y, Z, W; X', Y', Z', W'$ denote the differential coefficients of U, U' with respect to x, y, z, w ; then it is easy to see that the equation of the Intersect-Developable is obtained by eliminating ξ, η, ζ, ω between the equations

$$\begin{aligned} T &= 0, & T' &= 0, \\ X\xi + Y\eta + Z\zeta + W\omega &= 0, & X'\xi + Y'\eta + Z'\zeta + W'\omega &= 0. \end{aligned}$$

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If, for shortness, we suppose

$$\begin{aligned} F &= YZ' - Y'Z, & L &= XW' - X'W, \\ G &= ZX' - Z'X, & M &= YW' - Y'W, \\ H &= XY' - X'Y, & N &= ZW' - Z'W, \end{aligned}$$

(values which give rise to the identical equation $LF + MG + NH = 0$), then, λ, μ, ν, ρ denoting any indeterminate quantities, the two linear equations in ξ, η, ζ, ω are identically satisfied by assuming

$$\begin{aligned} \xi &= N\mu - M\nu + F\rho, \\ \eta &= -N\lambda + L\nu + G\rho, \\ \zeta &= M\lambda - L\mu + H\rho, \\ \omega &= -F\lambda - G\mu - H\nu. \end{aligned}$$

and, substituting these values in the equations $T = 0$, $T' = 0$, we have two equations:

$$\begin{aligned} A\lambda^2 + B\mu^2 + C\nu^2 + 2F\mu\nu + 2G\nu\lambda + 2H\lambda\mu + 2L\lambda\rho + 2M\mu\rho + 2N\nu\rho &= 0, \\ A'\lambda^2 + B'\mu^2 + C'\nu^2 + 2F'\mu\nu + 2G'\nu\lambda + 2H'\lambda\mu + 2L'\lambda\rho + 2M'\mu\rho + 2N'\nu\rho &= 0, \end{aligned}$$

which are of course such as to permit the four quantities λ, μ, ν, ρ to be simultaneously eliminated. The coefficients of these equations are obviously of the fourth order in x, y, z, w .

Suppose for a moment that these coefficients (instead of being such as to permit this simultaneous elimination of λ, μ, ν, ρ) denoted any arbitrary quantities, and suppose that the indeterminates λ, μ, ν, ρ were besides connected by two linear equations,

$$a\lambda + b\mu + c\nu + d\rho = 0, \quad a'\lambda + b'\mu + c'\nu + d'\rho = 0;$$

then, putting

$$\begin{aligned} bc' - b'c &= f, & ad' - a'd &= l, \\ ca' - c'a &= g, & bd' - b'd &= m, \\ ab' - a'b &= h, & cd' - c'd &= n, \end{aligned}$$

(values which give rise to the identical equation $lf + mg + nh = 0$), and effecting the elimination of λ, μ, ν, ρ between the four equations, we should obtain a final equation $\square = 0$, in which $\square$ is a homogeneous function of the second order in each of the systems of coefficients $A, B, \&c.$ and $A', B', \&c.$, and a homogeneous function of the fourth order (indeterminate to a certain extent in its form on account of the identical

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equation $lf + mg + nh = 0$) in the coefficients f, g, h, l, m, n .⁴ But re-establishing the actual values of the coefficients $A, B, \&c.$, $A', B', \&c.$ (by which means the function $\square$ becomes a function of the sixteenth order

⁴I believe the result of the elimination is

$$\square = \left(\frac{QR' - Q'R}{f} \right)^2 - 4 \left(\frac{PR' - P'R}{g} \right) \left(\frac{PQ' - P'Q}{h} \right),$$

in x, y, z, w) the quantities f, g, h, l, m, n ought, it is clear, to disappear of themselves; and the way this happens is that the function $\square$ resolves itself into the product of two factors M and Φ , the latter of which is independent of f, g, h, l, m, n . The factor M is a function of the fourth order in these quantities, and it is also of the eighth order in the variables x, y, z, w , the factor Φ is consequently of the eighth order in x, y, z, w . And the result of the elimination being represented by the equation $\Phi = 0$, the Intersect-Developable in the general case, or (what is the same thing) for systems of the class A, is of the eighth order. In the case of a system of the class B the equation obtained as above contains as a factor the square, and in the case of a system of the class C the cube, of the linear function which equated to zero is the equation of the plane of contact. The Intersect-Developable of a system of the class B is therefore a Developable of the sixth order, and that of a system of the class C a Developable of the fifth order. The elimination is in every case most simply effected by supposing two of the quantities λ, μ, ν, ρ to vanish (e.g. ν and ρ): the equations between which the elimination has to be effected then are

$$A\lambda^2 + B\mu^2 + 2H\lambda\mu = 0, \quad A'\lambda^2 + B'\mu^2 + 2H'\lambda\mu = 0;$$

and the result may be presented under the equivalent forms

$$(AB' + A'B - 2HH')^2 - 4(AB - H^2)(A'B' - H'^2) = 0,$$

and

$$(AB' - A'B)^2 + 4(AH' - A'H)(BH' - B'H) = 0,$$

the latter of which is the most convenient. These two forms still contain an extraneous factor of the eighth order in x, y, z, w , of which they can only be divested by substituting the actual values of A, B, H, A', B', H' .

A. Two surfaces forming a system belonging to this class may be represented by equations of the form

$$ax^2 + by^2 + cz^2 + dw^2 = 0, \quad a'x^2 + b'y^2 + c'z^2 + d'w^2 = 0,$$

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where $x = 0, y = 0, z = 0$, and $w = 0$, are the equations of the four conjugate planes. There is no particular difficulty in performing the operations indicated by the general process given above; and if we write, in order to abbreviate,

$$\begin{aligned} bc' - b'c &= f, & ad' - a'd &= l, \\ ca' - c'a &= g, & bd' - b'd &= m, \\ ab' - a'b &= h, & cd' - c'd &= n, \end{aligned}$$

(values which satisfy the identical equation $lf + mg + nh = 0$), the result after all reductions is

$$\begin{aligned} & l^2 f^2 y^2 z^2 + m^2 g^2 z^2 x^2 + n^2 h^2 x^2 y^2 + l^2 f^2 x^2 w^2 + m^2 g^2 y^2 w^2 + n^2 h^2 z^2 w^2 \\ & + 2mnfh y^2 z^2 + 2nlhg z^2 x^2 + 2lmfg x^2 y^2 - 2m^2 ngh w^2 yz - 2n^2 lhf w^2 zx - 2l^2 mfg w^2 xy \\ & + 2fmg^2 x^2 zw + 2gnh^2 y^2 xw + 2hlf^2 z^2 yw - 2fnh^2 x^2 yw - 2glf^2 y^2 zw - 2hmg^2 z^2 xw \\ & + 2(mg - nh)(nh - lf)(lf - mg)x^2 y^2 z^2 w^2 = 0, \end{aligned}$$

which is therefore the equation of the Intersect-Developable for this case. The discussion of the geometrical properties of the surface will be very much facilitated by presenting the equation under the following form, which is evidently one of a system of six different forms,

$$\begin{aligned} & \{m(gx^2 + nw^2)(hy^2 - gz^2 + lw^2) - l(-fx^2 + nw^2)(-hx^2 + fz^2 + mw^2)\}^2 \\ & - 4fglm x^2 y^2 (hy^2 - gz^2 + lw^2)(-hx^2 + fz^2 + mw^2) = 0. \end{aligned}$$

where, if we write $uA + u'A' = A$, &c., the quantities P, Q, R are given by the equation (identical with respect to u, u')

$$\begin{aligned} Pu^2 + 2Quu' + Ru'^2 &= (Aa^2 + \dots)(\Phi l^2 + \dots) - (Aal + \dots)^2 \\ &= u^2\{(BC - F^2)f^2 + \dots\} + uu'\{(BC' + B'C - 2FF')f^2 + \dots\} \\ &\quad + u'^2\{(B'C' - F'^2)f^2 + \dots\}. \end{aligned}$$

A theorem connected with that given in the second part of my memoir 'On Linear Transformations' (*Journal*, vol. I. p. 109) [see [14], p. 100]. I am not in possession of any verification *a posteriori* of what is subsequently stated as to the resolution into factors of the function $\square$ and the forms of these factors.

B. Two surfaces forming a system belonging to this class may be represented by equations such as

$$ax^2 + by^2 + cz^2 + 2nzw = 0, \quad a'x^2 + b'y^2 + c'z^2 + 2n'zw = 0,$$

in which $x = 0$, $y = 0$ are the equations of the single conjugate planes, $z = 0$ that of the double conjugate plane or plane of contact, $w = 0$ that of an indeterminate plane through the two single conjugate points. If we write

$$\begin{aligned} bc' - b'c &= f, & an' - a'n &= p, \\ ca' - c'a &= g, & bn' - b'n &= q, \\ ab' - a'b &= h, & cn' - c'n &= r, \end{aligned}$$

(values which satisfy the identical equation $pf + qg + rh = 0$), the result after all reductions is

$$\begin{aligned} r^2h^2z^4 + 2pr^2h(rh - qg)z^3x - 2qr^2h(pf - rh)z^3y + 4fqr^2hz^2x^2w - 4pqr^2hz^2y^2w \\ + p^2(rh - qg)^2z^2x^2 + q^2(pf - rh)^2z^2y^2 + 2pq(4r^2h^2 - fgpq)z^2xy \\ + 4p^2q(rh - qg)zx^2w + 4pq^2(pf - rh)zy^2w - 4p^2q^2(qg - pf)zxyw \\ + 4p^4q^2x^4w^2 + 4p^2q^4y^4w^2 + 8p^3q^3x^2y^2w^2 + 4p^2q^2h^2x^2y^2 = 0, \end{aligned}$$

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which is therefore the equation of the Intersect-Developable for systems of the case in question. The equation may also be presented under the form

$$\begin{aligned} \{q(px^2 + rz^2)(hy^2 - gz^2 + 2pzw) - p(qy^2 + rz^2)(-hx^2 + fz^2 + 2qzw)\}^2 \\ + 4pqx^2y^2(hy^2 - gz^2 + 2pzw)(-hx^2 + fz^2 + 2qzw) = 0, \end{aligned}$$

which it is to be remarked contains the extraneous factor z^2 . The following is also a form of the same equation,

$$\begin{aligned} r^2(qg - pf)z^4 - fr^2z^3x^2 + gr^2z^3y^2 + 2pq(px^2 + qy^2)w^2 \\ - 4pq(px^2 + qy^2 + rz^2)\{r(hy^2 - gz^2)(fz^2 - hx^2) + 2pq(gx^2 - fy^2)zw\} = 0. \end{aligned}$$

C. Two surfaces forming a system belonging to this class may be represented by equations of the form

$$ax^2 + by^2 + 2fyz + 2nzw = 0, \quad a'x^2 + b'y^2 + 2f'yz + 2n'zw = 0,$$

in which $bn' - b'n = 0$, $af' - a'f = 0$. In these equations $x = 0$ is the equation of a properly chosen plane passing through the two conjugate points, $y = 0$ is the equation of the single conjugate plane, $z = 0$ that of the triple conjugate plane, and $w = 0$ is the equation of a properly chosen plane passing through the single conjugate point. Or without loss of generality, we may write

$$\alpha(x^2 - 2yz) + \beta(y^2 - 2zw) = 0, \quad \alpha'(x^2 - 2yz) + \beta'(y^2 - 2zw) = 0,$$

where x, y, z and w have the same signification as before.⁵ The result after all reductions is

$$4x^3w + 12xy^2w + 9y^4z - 24x^2yzw - 4x^2y^3 + 8x^4 = 0,$$

⁵Of course in working out the equation of the Intersect-Developable, it is simpler to employ the equations $x^2 - 2yz = 0$, $y^2 - 2zw = 0$. These equations belong to two cones which pass through the Intersect and have their vertices in the triple conjugate point and single conjugate point respectively. I have not alluded to these cones in the text, as the theory of them does not come within the plan of the present memoir, the immediate object of which is to exhibit the equations of certain developable surfaces; but these cones are convenient in the present case as furnishing the easiest means of defining the planes $x = 0$, $w = 0$. If we represent for a moment the single conjugate point by S and the triple conjugate point by T (and the cones through these points by the same letters), then the point T is a point upon the cone S , and the triple conjugate plane which touches the cone S along the line TS touches the cone T along some generating line TM . Let the other tangent plane through the line TS to the cone T be TM' , where M' may represent the point where the generating line in question meets the cone S ; and we may consider M as the point of intersection of the line TM with the tangent plane through the line SM' to the cone S : then the plane TMM' is the plane represented by the equation $x = 0$, and the plane SMM' is that represented by the equation $w = 0$. We may add that $y = 0$ is the equation of the plane TSM' , and $z = 0$ that of the plane TSM .

which may also be presented under the forms

$$z(y^2 - 2zw)^2 - 4y(y^2 - 2zw)(x^2 - 2yz) + 8w(x^2 - 2yz)^2 = 0,$$

and

$$z(3y^2 + 2zw)^2 - 4x^2(y^2 - 2xw + 6yzw) = 0.$$

[In these three equations and in the last two equations of p. 495 as originally printed, there was by mistake, an interchange of the letters x and y .]

[PDF p. 514 / Book p. 492]

Proceeding next to the problem of finding the envelope of two surfaces of the second order, this is most readily effected by the following method communicated to me by Mr Salmon. Retaining the preceding notation, the equation $U + kU' = 0$ belongs to a surface of the second order passing through the Intersect of the two surfaces $U = 0$, $U' = 0$. The polar reciprocal of this surface $U + kU' = 0$ is therefore a surface inscribed in the envelope of the reciprocals of the two surfaces $U = 0$, $U' = 0$, and consequently this envelope is the envelope (in the ordinary sense of the word) of the reciprocal of the surface $U + kU' = 0$, k being considered as a variable parameter. It is easily seen that the reciprocal of the surface $U + kU' = 0$ is given by an equation of the form

$$A + 3Bk + 3Ck^2 + Dk^3 = 0,$$

in which A, B, C, D are homogeneous functions of the second order in the coordinates x, y, z, w . Differentiating with respect to k , and performing the elimination, we have for the equation of the envelope in question,

$$(AD - BC)^2 - 4(AC - B^2)(BD - C^2) = 0;$$

or the envelope is, in general or (what is the same thing) for a system of the class A, a developable of the eighth order. For a system of the class B the equation contains as a factor, the square of the linear function which equated to zero is the equation of the plane of contact; or the envelope is in this case a Developable of the sixth order. And in the case of a system of the class C the equation contains as a factor the cube of this linear function; or the envelope is a developable of the fifth order only.

A. We may take for the two surfaces the reciprocals (with respect to $x^2 + y^2 + z^2 + w^2 = 0$) of the equations made use of in determining the Intersect-Developable. The equations of these reciprocals are

$$bcdx^2 + cday^2 + dabz^2 + abcw^2 = 0, \quad b'c'd'x^2 + c'd'a'y^2 + d'a'b'z^2 + a'b'c'w^2 = 0;$$

and it is clear from the form of them (as compared with the equations of the surfaces of which they are the reciprocals) that $x = 0, y = 0, z = 0, w = 0$, are still the equations of the conjugate planes. We have, introducing the numerical factor 3 to avoid fractions,

$$\begin{aligned} & 3\{(b + kb')(c + kc')(d + kd')x^2 + (c + kc')(d + kd')(a + ka')y^2 \\ & + (d + kd')(a + ka')(b + kb')z^2 + (a + ka')(b + kb')(c + kc')w^2\} \\ & = A + 3Bk + 3Ck^2 + Dk^3, \end{aligned}$$

which determine the values of A, B, C, D . We have in fact

$$\begin{aligned} A &= 3(bcdx^2 + cday^2 + dabz^2 + abcw^2), \\ B &= (b'cd + bc'd + bcd')x^2 + \dots, \\ C &= (bc'd' + b'cd' + b'c'd)x^2 + \dots, \\ D &= 3(b'c'd'x^2 + c'd'a'y^2 + d'a'b'z^2 + a'b'c'w^2). \end{aligned}$$

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and these values give (with the same signification as before of f, g, h, l, m, n)

$$\begin{aligned} 2(AC - B^2) &= Aa^2 + Bb^2 + Cc^2 + 2Fbc + 2Gca + 2Hab + 2Lad + 2Mbd + 2Ncd + Dd^2, \\ 2(BD - C^2) &= Aa'^2 + Bb'^2 + Cc'^2 + 2Fb'c' + 2Gc'a' + 2Ha'b' \\ &\quad + 2La'd' + 2Mb'd' + 2Nc'd' + Dd'^2, \\ AD - BC &= Aaa' + Bbb' + Ccc' + F(bc' + b'c) + G(ca' + c'a) + H(ab' + a'b) \\ &\quad + L(ad' + a'd) + M(bd' + b'd) + N(cd' + c'd) + Ddd', \end{aligned}$$

where

$$\begin{aligned} A &= n^2y^2z^2 + m^2z^2x^2 - mn yz x^2 + 2lmnz^2w^2 - 2lmy^2w^2 + 2nmy^2z^2, & F &= l^2y^2z^2, \\ B &= l^2z^2x^2 + n^2x^2y^2 + ghw^4 + 2ghnx^2w^2 - 2glz^2w^2 + 2lnz^2x^2, & G &= m^2z^2x^2, \\ C &= m^2x^2y^2 + l^2y^2z^2 + f^2y^4 + 2lfy^2w^2 - 2hmx^2w^2 + 2mlx^2y^2, & H &= f^2x^2y^2, \\ D &= f^2w^4 + g^2y^4 + h^2z^4 - 2ghy^2z^2 - 2hfx^2z^2 - 2fgx^2y^2, & L &= f^2x^2w^2, \\ & & M &= g^2y^2w^2, \\ & & N &= h^2z^2w^2. \end{aligned}$$

and then $4(AC - B^2)(BD - C^2) - (AD - BC)^2 =$

$$\begin{aligned} &(BC - F^2)f^2 + (CA - G^2)g^2 + (AB - H^2)h^2 + (AD - L^2)l^2 + (BD - M^2)m^2 + (CD - N^2)n^2 \\ &+ 2(GH - AF)gh + 2(HF - BG)hf + 2(FG - CH)fg - 2(MN - DF)mn - 2(NL - DG)nl - 2(LM - DH)lm \\ &+ 2(AM - LH)lh + 2(BN - MF)mf + 2(CL - NG)ng - 2(AN - LG)lg - 2(BL - MH)mh - 2(CM - NF)nf \\ &+ 2(NH - MG)lf + 2(LF - NH)mg + 2(MG - LF)nh. \end{aligned}$$

Substituting the values of A, B , &c. in this expression, the result after all reductions is

$$\begin{aligned} &f^2m^2l^2x^2 + g^2n^2m^2y^2 + h^2l^2n^2z^2 + f^2g^2klhw^2 + 2g^2ln(mg - nh)y^2z^2 \\ &+ 2hm^2l(nh - lf)z^2x^2 + 2fn^2m(lf - mg)x^2y^2 - 2hl^2m(mg - nh)y^2w^2 \\ &- 2fm^2l(nh - lf)z^2w^2 - 2gn^2l(lf - mg)x^2w^2 + 2f^2mn(mg - nh)x^2zw \\ &+ 2g^2nl(nh - lf)y^2xw + 2h^2lm(lf - mg)z^2yw - 2f^2gh(mg - nh)x^2wy \\ &- 2fg^2h(nh - lf)y^2wz - 2fgh^2(lf - mg)z^2wx + f^2(l^2f^2 - 6ghmn)w^2x^2 \\ &+ g^2(m^2g^2 - 6hfnl)w^2y^2 + h^2(n^2h^2 - 6lmfg)w^2z^2 \\ &+ l^2(l^2f^2 - 6ghmn)y^2z^2 + m^2(m^2g^2 - 6hfnl)z^2x^2n^2(n^2h^2 - 6lmfg)x^2y^2 \\ &+ 2gh(ghmn - 3f^2l^2)w^2y^2z^2 + 2hf(hfnl - 3g^2m^2)w^2z^2x^2 \\ &+ 2fg(fglm - 3h^2n^2)w^2x^2y^2 + 2hm(ghmn - 3f^2l^2)z^2x^2w^2 \\ &+ 2fn(hfnl - 3g^2m^2)x^2y^2w^2 + 2gl(fglm - 3h^2n^2)y^2z^2w^2 \\ &- 2gn(ghmn - 3f^2l^2)y^2x^2w^2 - 2hl(hfnl - 3g^2m^2)z^2y^2w^2 \\ &- 2fm(fglm - 3h^2n^2)x^2z^2w^2 - 2mn(ghmn - 3f^2l^2)x^2y^2z^2 \\ &- 2nl(hfnl - 3g^2m^2)y^2z^2x^2 - 2lm(fglm - 3h^2n^2)z^2x^2y^2 \\ &- 2(mg - nh)(nh - lf)(lf - mg)x^2y^2z^2w^2 = 0, \end{aligned}$$

which is therefore the equation of the envelope for this case. The equation may also be presented under the form

$$u^2 + (mnf x + nlg y + lmh z)^2 \{f^2 w^2 + g^2 y^2 + h^2 z^2 - 2ghyz - 2hfzx - 2fgxy\} = 0;$$

and there are probably other forms proper to exhibit the different geometrical properties of the surface, but with which I am not yet acquainted.

B. Here taking for the two surfaces the reciprocals of the equations made use of in determining the Intersect-Developable, the equations of these reciprocals are

$$n^2 b x^2 + n^2 a y^2 - abc w^2 + 2 nab z w = 0, \quad n'^2 b' x^2 + n'^2 a' y^2 - a' b' c' w^2 + 2 n' a' b' z w = 0,$$

which are similar to the equations of the surfaces of which they are reciprocal, only z and w are interchanged, so that here $x = 0, y = 0$ are the single conjugate planes, $z = 0$ is an indeterminate plane passing through the single conjugate points, and $w = 0$ is the equation of the double conjugate plane or plane of contact.

The values of A, B, C, D are

$$\begin{aligned} A &= 3(n^2 b x^2 + n^2 a y^2 - abc w^2 + 2 nab z w), \\ B &= (2nn'b + n^2 b')x^2 + \dots, \\ C &= (2nn'b' + n'^2 b)x^2 + \dots, \\ D &= 3(n'^2 b' x^2 + n'^2 a' y^2 - a' b' c' w^2 + 2 n' a' b' z w). \end{aligned}$$

Hence, using f, g, h, p, q, r in the same sense as before, we have for $2(AC - B^2)$, $2(BD - C^2)$, $(AD - BC)$ expressions of the same form as in the last case (p, q, r being written for l, m, n), but in which

$$\begin{aligned} A &= f^2 w^4 + 4fhzwy + 8qry^2 w^2 - 4fqzw^3, & G &= -g^2 x^2, \\ B &= ghw^4 + 4gfhzxw + 8prx^2 w^2 + 4gpzw^3, & H &= 0, \\ C &= h^2 w^4, \\ D &= 2(fx^2 + gy^2)y^2 + 4h^2 z^2 w^2 + h^2 x^2 y^2 - 8qh x^2 zw + 8phy^2 zw \\ &\quad + 4ghy^2 w^2 + 2fhx^2 w^2, \\ L &= 2pqy^2 zw, & M &= 2pqx^2 zw, & N &= -2h^2 zw^2. \end{aligned}$$

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The substitution of these values gives after all reductions the result

$$\begin{aligned} &f^2 g^2 h^2 w^4 + 4(pf - qg) fgh^2 zw^3 + 4(r^2 h^2 - 6pqgf) h^2 z^2 w^2 \\ &+ 2(q^2 r^2 + 2prfh) fhx^2 y^2 w^2 + 2(p^2 r^2 + 2qrgh) ghy^2 x^2 w^2 \\ &- 16(pf - qg)^2 w^4 - 4(q^2 r^2 - 4p^2 r^2 - 6pqfg) qhx^2 zw^2 \\ &- 4(p^2 r^2 - 4q^2 r^2 - 6pqfg) phy^2 zw^2 + 16p^2 q^2 h^2 z^4 w^2 \\ &- 8(pf + 4qg) q^2 phx^2 z^2 w^2 - 8(qg + 4pf) pq^2 hy^2 z^2 w^2 \\ &+ (q^2 r^2 + 8prfh) q^2 x^4 w^2 + (p^2 r^2 + 8qrgh) p^2 y^4 w^2 \\ &+ 2(10r^2 h^2 - pqfg) pqx^2 y^2 w^2 - 16pr^2 hx^2 z^2 w + 16qr^2 hy^2 z^2 w \\ &+ 4(4pf + 9qg) pq^2 x^4 zw - 4(4qg + 9pf) p^2 qy^4 zw - 4(pf - qg) p^2 q^2 x^2 y^2 zw \\ &+ 4p^2 q^4 x^2 z^2 + 4p^4 q^2 y^2 z^2 + 8p^3 q^3 x^2 y^2 z^2 + 4pq^4 rx^4 + 4p^4 qry^4 \\ &+ 12p^3 q^2 rx^2 y + 12p^2 q^3 rxy^2 = 0, \end{aligned}$$

which is therefore the equation of the envelope for this case. This equation may be presented under the form

$$w^4 + 4pq(qx^2 + py^2)(qrx^2 + rpy^2 + pqz^2) = 0,$$

and there are probably other forms which I am not yet acquainted with.

C. The reciprocals of the two surfaces made use of in determining the Intersect-Developable, although in reality a system of the same nature with the surfaces of which they are reciprocals, are represented by equations of a somewhat different form. There is no real loss of generality in replacing the two surfaces by the reciprocals of the cones $x^2 = 2yz$, $y^2 = 2zw$; or we may take the two conies

$$(x^2 - 2yz = 0, w = 0) \quad \text{and} \quad (y^2 - 2zw = 0, x = 0),$$

for the surfaces of which the envelope has to be found, these conies being, it is evident, the sections by the planes $w = 0$ and $x = 0$ respectively of the cones the Intersect-Developable of which was before determined. The process of determining the envelope is however essentially different: supposing the plane $\xi x + \eta y + \zeta z + \omega w = 0$ to be the equation of a tangent plane to the two conies (that is, of a plane passing through a tangent of each of the conies) then the condition of touching the first conic gives $\xi^2 - 2\eta\zeta = 0$, and that of touching the second conic gives $\eta^2 - 2\zeta\omega = 0$. We have therefore to find the envelope (in the ordinary sense of the word) of the plane $\xi x + \eta y + \zeta z + \omega w = 0$, in which the coefficients ξ, η, ζ, ω are variable quantities subject to the conditions

$$\xi^2 - 2\eta\zeta = 0, \quad \eta^2 - 2\zeta\omega = 0.$$

The result which is obtained without difficulty by the method of indeterminate multipliers, [or more easily by writing $\xi : \eta : \zeta : \omega = 2\theta^3 : 2\theta^2 : \theta : 2$] is

$$8y^3z - 32y^2z^2w + 32z^3w^2 - 27x^2w + 27xyzw - 4x^3y = 0,$$

which may also be written under the form

$$8z(y^2 - 2zw)^2 - x^2\{4y^2 + 9(3x^2 - 8yz)w\} = 0.$$

[Another form, containing the factor w , is $4\{y^3 + 2zwy - (2y^2 + 27x^2w - 30yzw)\}^2 = 0$.]

[85] Note on a Family of Curves of the Fourth Order

[From the *Cambridge and Dublin Mathematical Journal*, vol. v. (1850), pp. 148–152.]

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The following theorem, in a slightly different and somewhat less general form, is demonstrated in Mr Hearn's "Researches on Curves of the Second Order, &c.", Lond. 1846: "The locus of the pole of a line, $u + v + w = 0$, with respect to the conics passing through the angles of the triangle ($u = 0, v = 0, w = 0$), and touching a fixed line $\alpha u + \beta v + \gamma w = 0$, is the curve of the fourth order,

$$\sqrt{\alpha u(v + w - u)} + \sqrt{\beta v(w + u - v)} + \sqrt{\gamma w(u + v - w)} = 0;$$

" the difference in fact being, that with Mr Hearn the indeterminate line $u + v + w = 0$ is replaced by the line ∞ , so that the poles in question become the centres of the conics.

Previous to discussing the curve of the fourth order, it will be convenient to notice a property of curves of the fourth order with three double points. Such curves contain eleven arbitrary constants: or if we consider the double points as given, then five arbitrary constants. From each double point may be drawn two tangents to the curve; any five of the points of contact of these tangents determine the curve, and consequently determine the sixth point of contact. The nature of this relation will be subsequently explained; at present it may be remarked that it is such that, if three of the points of contact (each one of such points of contact corresponding to a different double point) lie in a straight line, the remaining three points of contact also lie in a straight line. A curve of the fourth order having three given double points and besides such that the points of contact of the tangents from the double points lie three and three in two straight lines, contains therefore four arbitrary constants. Now it is easily seen that the curve in question has three double points, viz. the points given by the equations

$$(u = 0, v - w = 0), \quad (v = 0, w - u = 0), \quad (w = 0, u - v = 0),$$

points which may be geometrically defined as the projections from the angles of the triangle ($u = 0, v = 0, w = 0$) upon the opposite sides, of the point ($u = v = w$) which is the harmonic with respect to the triangle of the given line $u + v + w = 0$. Moreover,

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the lines $u = 0, v = 0, w = 0$ (being lines which, as we have seen, pass through the double points) touch the curve in three points lying in a line, viz. the given line $\alpha u + \beta v + \gamma w = 0$. Hence the curve in question is a curve with three double points, such that the points of contact of the tangents from the double points lie three and three in two straight lines. Considering the double points as given, the functions u, v, w contain two arbitrary ratios, and the ratios of the quantities α, β, γ being arbitrary, the equation of the curve contains four arbitrary constants, or it represents the general curve of the class to which it has been stated to belong.

As to the investigation of the above-mentioned theorem with respect to curves of the fourth order with three double points, the general form of the equation of such a curve is

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0,$$

where the double points are the angles of the triangle ($x = 0, y = 0, z = 0$). It may be remarked in passing, that the six tangents at the double point touch the conic

$$ax^2 + by^2 + cz^2 - 2fyz - 2gxz - 2hxy = 0.$$

To determine the tangents through ($y = 0, z = 0$), we have only to write the equation to the curve under the form

$$\frac{a}{x^2} + \frac{2h}{xy} + \frac{2g}{xz} + \frac{b}{y^2} + \frac{2f}{yz} + \frac{c}{z^2} = 0;$$

the points of contact are given by the system

$$\begin{aligned} Ax + By + Cz &= 0, \\ \frac{C}{y^2} + \frac{B}{z^2} + \frac{2F}{yz} &= 0, \end{aligned}$$

the latter equation (which evidently belongs to a pair of lines) determining the tangents. The former equation is that of a conic passing through the angles of the triangle $x = 0, y = 0, z = 0$: since the tangents pass through the point $y = 0, z = 0$ they evidently each intersect the conic in one other point only. The equation of the tangents shows that these lines are the tangents through the point $y = 0, z = 0$ to the conic whose equation is

$$aA^2x^2 + bB^2y^2 + cC^2z^2 + 2fBCyz + 2gCAzx + 2hABxy = 0.$$

To complete the construction of the points of contact it may be remarked, that the equations which determine the coordinates of these points may be presented under the form

$$\begin{aligned} Ax &= [A]x, \\ By &= [B + F(y/z)]y, \\ Cz &= [C + F(z/y)]z. \end{aligned}$$

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whence also

$$\begin{aligned} aAx + hBy + gCz &= Kx, \\ hAx + bBy + fCz &= G\sqrt{K}y, \\ gAx + fBy + cCz &= -H\sqrt{K}x. \end{aligned}$$

or writing for shortness ξ, η, ζ instead of the linear functions forming the first sides of these equations respectively, we have

$$\frac{\xi}{\sqrt{-aK}} + \frac{\eta}{G} + \frac{\zeta}{H} = 0,$$

from which it follows at once that the equation to the line forming the two points of contact is

$$\frac{\xi}{\sqrt{-aK}} + \frac{\eta}{G} + \frac{\zeta}{H} = 0;$$

and this may again be considered as the line joining the points $(\xi = 0, \eta/G + \zeta/H = 0)$ and $(\eta = 0, \zeta = 0)$.

Now $\xi = 0, \eta = 0, \zeta = 0$, are the polars of the double points (or angles of the triangle $x = 0, y = 0, z = 0$) with respect to the last-mentioned conic. The equation of the line joining the point $(y = 0, z = 0)$ with the point $(\eta = 0, \zeta = 0)$, is easily seen to be $GB y - CH z = 0$; from which it follows, that the lines forming each double point with the intersection of the polars of the other two double points meet in a point, the coordinates of which are given by

$$x : y : z = \frac{1}{AF} : \frac{1}{BG} : \frac{1}{CH},$$

and the polar of this point is the line

$$\frac{\xi}{F} + \frac{\eta}{G} + \frac{\zeta}{H} = 0.$$

The construction of this line is thus determined; and we have already seen how this leads to the construction of the lines joining the points of contact of the tangents from the double points. In fact, in the figure, if $\alpha\beta\gamma$ be the triangle whose sides are $\xi = 0, \eta = 0, \zeta = 0$, and A, B, C the points of intersection of the sides of this triangle with the line

$$\frac{\xi}{F} + \frac{\eta}{G} + \frac{\zeta}{H} = 0,$$

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the lines in question are $A\alpha, B\beta, C\gamma$. Moreover, the points of contact upon $A\alpha$ are harmonics with respect to A, α , and in like manner the points of contact on $B\beta, C\gamma$ are respectively harmonics with respect to B, β and to C, γ . This proves that if three of the points of contact are in the same straight line, the remaining three are also in the same straight line; in fact, we may consider three of the points of contact as given by

$$\begin{aligned}\xi : \eta : \zeta &= \sqrt{-aK} : G : H, \\ \xi : \eta : \zeta &= F : \sqrt{-bK} : H, \\ \xi : \eta : \zeta &= F : G : \sqrt{-cK},\end{aligned}$$

and the condition that these may be in the same line is

$$\sqrt{-a}\sqrt{-b}\sqrt{-c} + GH\sqrt{-a} + HF\sqrt{-b} + FG\sqrt{-c} = 0,$$

which remains unaltered when the signs of all the roots are changed. The equation just obtained may be considered as the condition which must exist between the coefficients of

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$$

in order that this curve may be transformable into the form

$$\sqrt{\alpha u(v+w-u)} + \sqrt{\beta v(w+u-v)} + \sqrt{\gamma w(u+v-w)} = 0.$$

[86] On the Developable derived from an Equation of the Fifth Order

[From the *Cambridge and Dublin Mathematical Journal*, vol. v. (1850), pp. 152–159.]

[PDF p. 522 / Book p. 500]

Möbius, in his “Barycentrische Calcul,” [Leipzig, 1827], has considered, or rather suggested for consideration, the family of curves of double curvature given by equations such as $x : y : z : w = A : B : C : D$, where A, B, C, D are rational and integral functions of an indeterminate quantity t . Observing that the plane $Ax + By + Cz + Dw = 0$ may be considered as the polar of the point determined by the system of equations last preceding, the reciprocal of the curve of Möbius is the developable, which is the envelope of a plane the coefficients in the equation of which are rational and integral functions of an indeterminate

quantity t , or what is equivalent, homogeneous functions of two variables ξ, η . Such an equation may be represented by $U = a\xi^n + nb\xi^{n-1}\eta + \dots = 0$, (where $a, b, \&c.$ are linear functions of the coordinates); and we are thus led to the developables noticed, I believe for the first time, in my “Note sur les Hyper-determinants,” Crelle, t. xxxiv. p. 148, [54]. I there remarked, that not only the equation of the developable was to be obtained by eliminating ξ, η from the first derived equations of $U = 0$; but that the second derived equations conducted in like manner to the edge of regression, and the third derived equations to the cusps or stationary points of the edge of regression. It followed that the order of the surface was $2(n-1)$, that of the edge of regression $3(n-2)$, and the number of stationary points $4(n-3)$. These values lead at once, as Mr Salmon pointed out to me, to the table,

$$\begin{aligned} m &= 3(n-2), \\ n &= n, \\ r &= 2(n-1), \\ a &= 0, \\ \beta &= 4(n-3), \\ \gamma &= \frac{1}{2}(n-1)(n-2), \\ h &= \frac{1}{2}(9n^2 - 53n + 80), \\ x &= 2(n-2)(n-3), \\ y &= 2(n-1)(n-3). \end{aligned}$$

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where the letters in the first column have the same signification as in my memoir in Liouville, [30], translated in the last number of the Journal. The order of the nodal line is of course $2(n-2)(n-3)$; Mr Salmon has ascertained that there are upon this line $6(n-3)(n-4)$ stationary points and $\frac{3}{4}(n-3)(n-4)(n-5)$ real double points, (the stationary points lying on the edge of regression, and with the stationary points of the edge of regression forming the system of intersections of the nodal line and edge of regression, and the real double points being triple points upon the surface). Also, that the number of apparent double points of the nodal line is

$$(n-3)(2n^3 - 18n^2 + 57n - 65).$$

The case of U a function of the second order gives rise to the cone $ac - b^2 = 0$. When U is a function of the third order, we have the developable

$$4(ac - b^2)(bd - c^2) - (ad - bc)^2 = 0,$$

which is the general developable of the fourth order having for its edge of regression the curve of the third order,

$$ac - b^2 = 0, \quad bd - c^2 = 0, \quad ad - bc = 0,$$

which is likewise the most general curve of this order: there are of course in this case no stationary points on the edge of regression. In the case where U is of the fourth order we have the developable of the sixth order,

$$(ae - 4bd + 3c^2)^3 - 27(ace + 2bcd - ad^2 - b^2e - c^3)^2 = 0;$$

having for its edge of regression the curve of the sixth order,

$$ae - 4bd + 3c^2 = 0, \quad ace + 2bcd - ad^2 - b^2e - c^3 = 0,$$

with four stationary points determined by the equations

$$\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e}.$$

The form exhibiting the nodal line of the surface has been given in the Journal by Mr Salmon. I do not notice it here, but pass on to the principal subject of the present paper, which is to exhibit the edge of regression and the stationary points of this edge of regression for the developable obtained from the equation of the fifth order,

$$U = a\xi^5 + 5b\xi^4\eta + 10c\xi^3\eta^2 + 10d\xi^2\eta^3 + 5e\xi\eta^4 + f\eta^5 = 0;$$

viz. that represented by the equation

$$\square = 0 = a^2 f^4 + 160a^3 c e f^2 + \cdots - 4000b^2 c^3 e^2,$$

[I do not reproduce here this expression for the discriminant of the binary quintic] a result for which I am indebted to Mr Salmon.

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To effect the reduction of this expression, consider in the first place the equations which determine the stationary points of the edge of regression. Writing instead of $\xi : \eta$ the single letter t , these equations are

$$at^2 + 2bt + c = 0,$$

$$bt^2 + 2ct + d = 0,$$

$$ct^2 + 2dt + e = 0,$$

$$dt^2 + 2et + f = 0,$$

write for shortness

$$A = 2(bf - 4ce + 3d^2), \quad B = af - 3be + 2cd, \quad C = 2(ae - 4bd + 3c^2),$$

and let $\alpha, 3\beta, 3\gamma, \delta$ represent the terms of

$$\begin{vmatrix} a & b & c & d \\ b & c & d & e \\ c & d & e & f \end{vmatrix},$$

viz.

$$\alpha = bdf - be^2 + 2cde - c^2f - d^3,$$

$$3\beta = adf - ae^2 - bcf + bde + d^2e - cd^2,$$

$$3\gamma = acf - ade - b^2f + bd^2 + bce - c^2d,$$

$$\delta = ace - ad^2 - b^2e + 2bcd - c^3;$$

it is obvious at first sight that the result of the elimination of t from the four quadratic equations is the system (equivalent of course to three equations),

$$\alpha = 0, \quad \beta = 0, \quad \gamma = 0, \quad \delta = 0.$$

The system in question may however be represented under the more simple form (which shows at once that the number of stationary points is, as it ought to be, eight),

$$A = 0, \quad B = 0, \quad C = 0;$$

this appears from the identical equations,

$$(2ct + 3d)(bt^2 + 2ct + d) - (2bt + 4c)(ct^2 + 2dt + e) + b(dt^2 + 2et + f) = \frac{1}{2}A,$$

$$(2ct + 3d)(at^2 + 2bt + c) - c(bt^2 + 2ct + d)$$

$$- (2at + 3b)(ct^2 + 2dt + e) + a(dt^2 + 2et + f) = B;$$

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$$(2bt + 3c)(at^2 + 2bt + c) - (2at + 4b)(bt^2 + 2ct + d) + a(ct^2 + 2dt + e) = \frac{1}{2}C,$$

(formulae the first and third of which are readily deduced from an equation given in the Note on Hyperdeterminants above quoted). The connexion between the quantities A, B, C and $\alpha, \beta, \gamma, \delta$, is given by

$$Aa - 2Bb + Cc = -6\delta,$$

$$Ab - 2Bc + Cd = -6\gamma,$$

$$Ac - 2Bd + Ce = -6\beta,$$

$$Ad - 2Be + Cf = -6\alpha.$$

The theory of the stationary points being thus obtained, the next question is that of finding the equations of the edge of regression. We have for this to eliminate t from the three cubic equations,

$$\begin{aligned} at^3 + 3bt^2 + 3ct + d &= 0, \\ bt^3 + 3ct^2 + 3dt + e &= 0, \\ ct^3 + 3dt^2 + 3et + f &= 0 : \end{aligned}$$

treating the quantities $t^3, t^2, t, 1$ as if they were independent, we at once obtain

$$\beta t + \alpha = 0, \quad \delta t + \gamma = 0, \quad \gamma t^2 - \alpha = 0, \quad \beta t^2 - \delta = 0;$$

or as this system may be more conveniently written,

$$\beta t + \alpha = 0, \quad \gamma t + \beta = 0, \quad \delta t + \gamma = 0.$$

But the most simple forms are obtained from the identical equations,

$$\begin{aligned} ft(at^3 + 3bt^2 + 3ct + d) - (3et + f)(bt^3 + 3ct^2 + 3dt + e) \\ + (2dt + e)(ct^3 + 3dt^2 + 3et + f) &= t^2(Bt + A); \\ (bt + c)(at^3 + 3bt^2 + 3ct + d) - (at + 3b)(bt^3 + 3ct^2 + 3dt + e) \\ + a(ct^3 + 3dt^2 + 3et + f) &= Ct + B, \end{aligned}$$

equations which, combined with those which precede, give the complete system

$$\beta t + \alpha = 0, \quad \gamma t + \beta = 0, \quad \delta t + \gamma = 0, \quad Bt + A = 0, \quad Ct + B = 0.$$